

Lab Program 02: Implement Vacuum Cleaner

#GOAL BASED AGENT

```
status = {}

for i in range(8):

    status['A'] = input("enter room A status clean or dirty ( C or D )").capitalize()
    status['B'] = input("enter room B status clean or dirty ( C or D )").capitalize()
    location = input("enter the location of vacuum cleaner A or B").capitalize()

    if(status['A']=='C' and status['B']=='C'):
        print("both rooms are clean")

    while(status['A']!='C' or status['B']!='C'):

        if(location == 'A' and status['A'] == 'C'):
            print("room A is clean so MOVE RIGHT")
            location = 'B'

        elif(location == 'A' and status['A'] == 'D'):
            print("room A is dirty so PICK THE DUST")
            status['A'] = 'C'

        elif(location == 'B' and status['B'] == 'D'):
            print("room B is dirty so PICK THE DUST")
            status['B'] = 'C'

        elif(location == 'B' and status['B'] == 'C'):
            print("room B is clean so MOVE RIGHT")
            location = 'A'

    else:
        print("ERROR")
```

Output:

```
IDLE Shell 3.12.8
File Edit Shell Debug Options Window Help
Python 3.12.8 (tags/v3.12.8:2dc476b, Dec 3 2024, 19:30:04) [MSC v.1942 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/admin/AppData/Local/Programs/Python/Python312/AI_183/Vaccum_simple reflex.py
enter room A status clean or dirty ( C or D )c
enter room B status clean or dirty ( C or D )d
enter the location of vaccum cleaner A or Ba
room A is clean so MOVE RIGHT
room B is dirty so PICK THE DUST
enter room A status clean or dirty ( C or D )d
enter room B status clean or dirty ( C or D )c
enter the location of vaccum cleaner A or Ba
room A is dirty so PICK THE DUST
enter room A status clean or dirty ( C or D )c
enter room B status clean or dirty ( C or D )c
enter the location of vaccum cleaner A or Ba
both rooms are clean
enter room A status clean or dirty ( C or D )d
enter room B status clean or dirty ( C or D )d
enter the location of vaccum cleaner A or Ba
room A is dirty so PICK THE DUST
room A is clean so MOVE RIGHT
room B is dirty so PICK THE DUST
enter room A status clean or dirty ( C or D )c
enter room B status clean or dirty ( C or D )d
enter the location of vaccum cleaner A or Bb
room B is dirty so PICK THE DUST
enter room A status clean or dirty ( C or D )d
enter room B status clean or dirty ( C or D )c
enter the location of vaccum cleaner A or Bb
room B is clean so MOVE RIGHT
room A is dirty so PICK THE DUST
enter room A status clean or dirty ( C or D )c
enter room B status clean or dirty ( C or D )c
enter the location of vaccum cleaner A or Bb
both rooms are clean
enter room A status clean or dirty ( C or D )d
enter room B status clean or dirty ( C or D )d
enter the location of vaccum cleaner A or Bb
room B is dirty so PICK THE DUST
room B is clean so MOVE RIGHT
room A is dirty so PICK THE DUST
```